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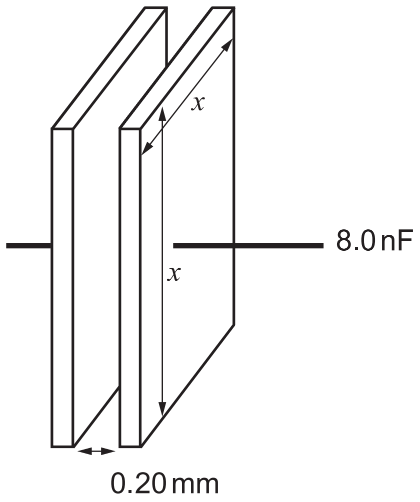
SECTION A

Answer **all** *questions.*

1. *(a)* (i) An 8.0 nF capacitor has square plates separated by 0.20 mm of air. Calculate the length (x) of the sides of the plates. [3]



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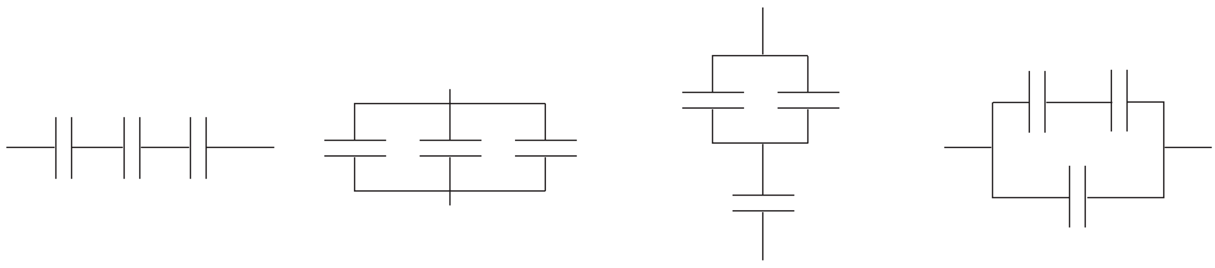
(ii) Which of the following combinations of 8.0 nF capacitors has a capacitance of 12.0 nF? Justify your answer. [4]

All individual capacitors shown are 8.0 nF.



1 2 3 4

(iii) State how the capacitance of the capacitor can be increased without changing its dimensions. [1]

(iv) Calculate the charge on the plates of the 8.0 nF capacitor when it stores an energy of 0.62 μJ. [3]

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Turn over.

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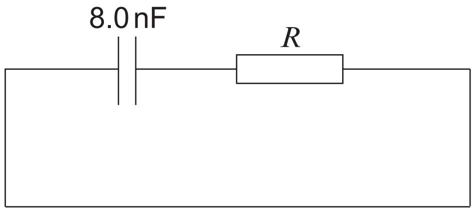
(b) (i) A charged 8.0 nF capacitor is discharged through a large resistor. Calculate the resistance of the resistor if the time constant is 15.5 ms. [2]



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(ii) Calculate the fraction of the initial charge remaining on the capacitor after 31.0 ms. [3]

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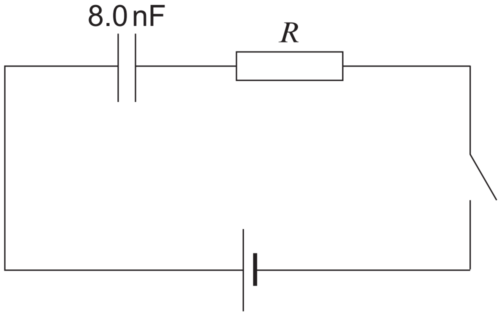
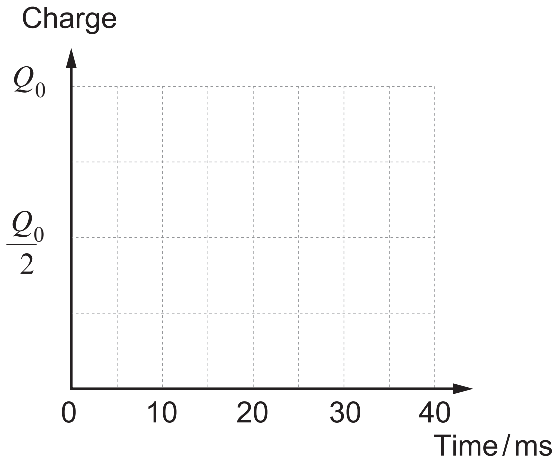
(c) An identical, uncharged 8.0 nF capacitor is **charged** in the following circuit, **using the same large resistor, *R***. The switch is closed and the capacitor is uncharged at time $t = 0$.

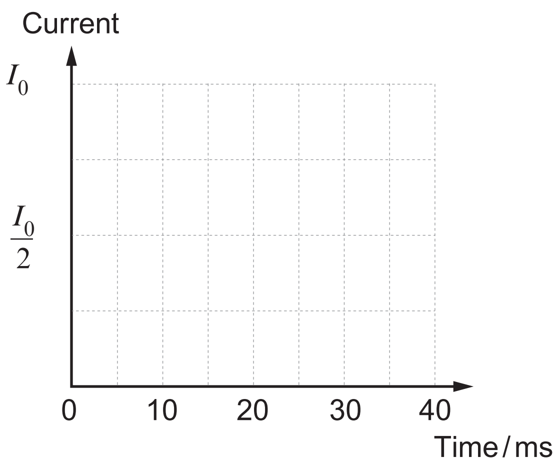


Without further calculations, sketch graphs of:

(i) the charge on the plates of the capacitor against time; [2]



(ii) the current in the circuit against time. [2]



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